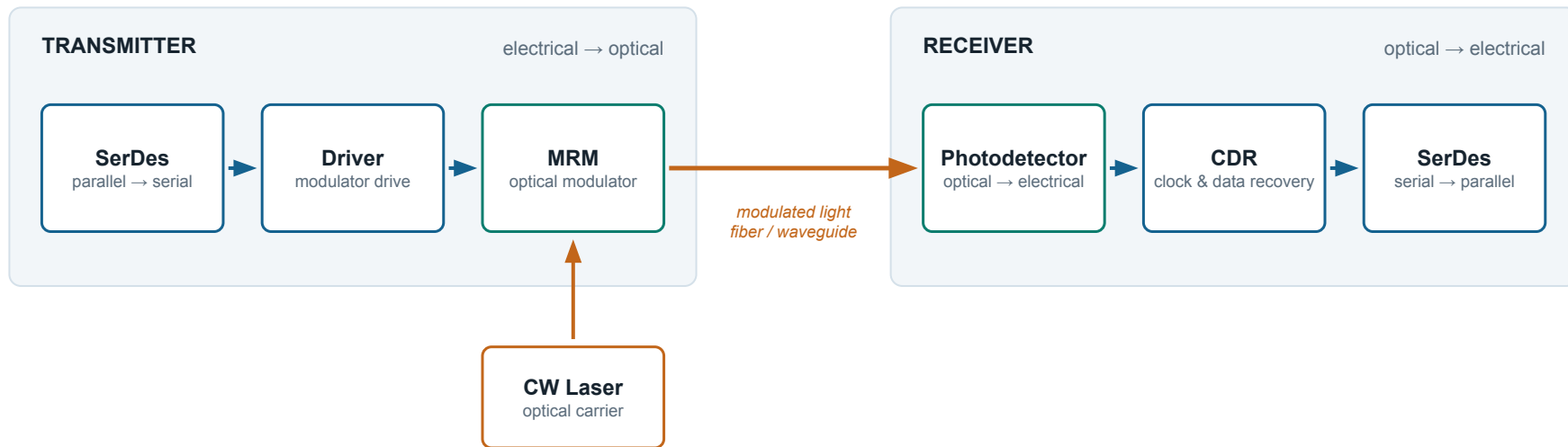


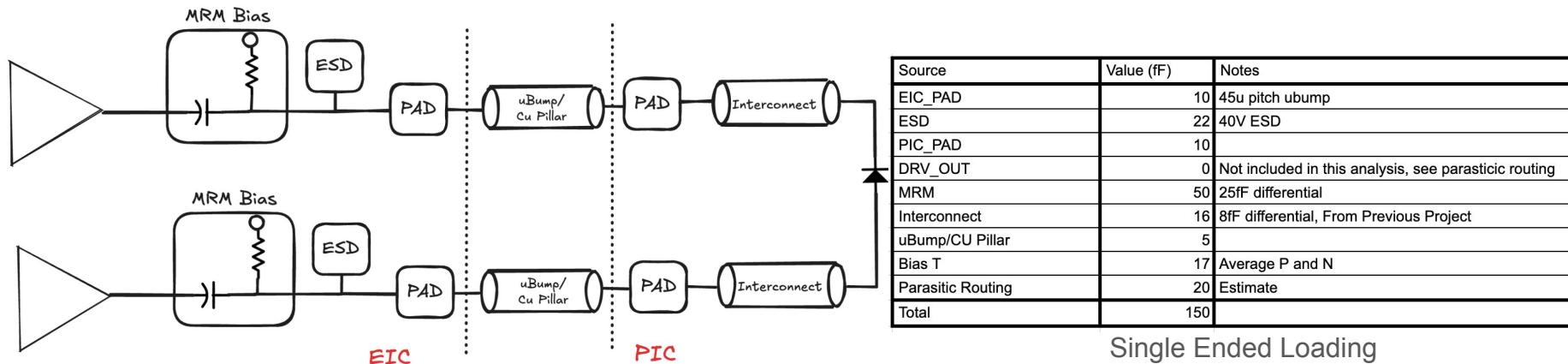
Optical Link Overview

Micro-ring modulator (MRM) based link — block level



- This presentation focuses on the Driver side.
- The MRM can be considered a PN junction from an electrical point of view.

The Loading Model and Calculation



* Note: The ESD requirements may be relaxed to 30V.

Driver Architecture Choice Matrix And Assumptions

Parameter	Assumption / Value	Design Considerations & Notes
Output Load	Lumped 150 fF	Realistic load offers slightly more relaxed condition due to MRM series resistance.
Output Swings	2V, 3V (pp diff)	Explored output swings for drivers.
Driver Output	Current Mode CML Voltage Mode	Key architecture candidates for design comparison.
Impedance Matching	No Explicit Matching	Relaxed effort due to the short channel between EIC and PIC.

Note: The simulations here are obtained under these conditions:

1. devices are extracted up to M5. The inter-device routing parasitics are not accounted for.
2. The core devices are assumed to be able to tolerate up to 0.96V.
3. The power consumption number is obtained from a 1.5V supply. If LDOs are used the numbers need to increase according to LDO drop-out.

Fundamental Limitations: Current Mode

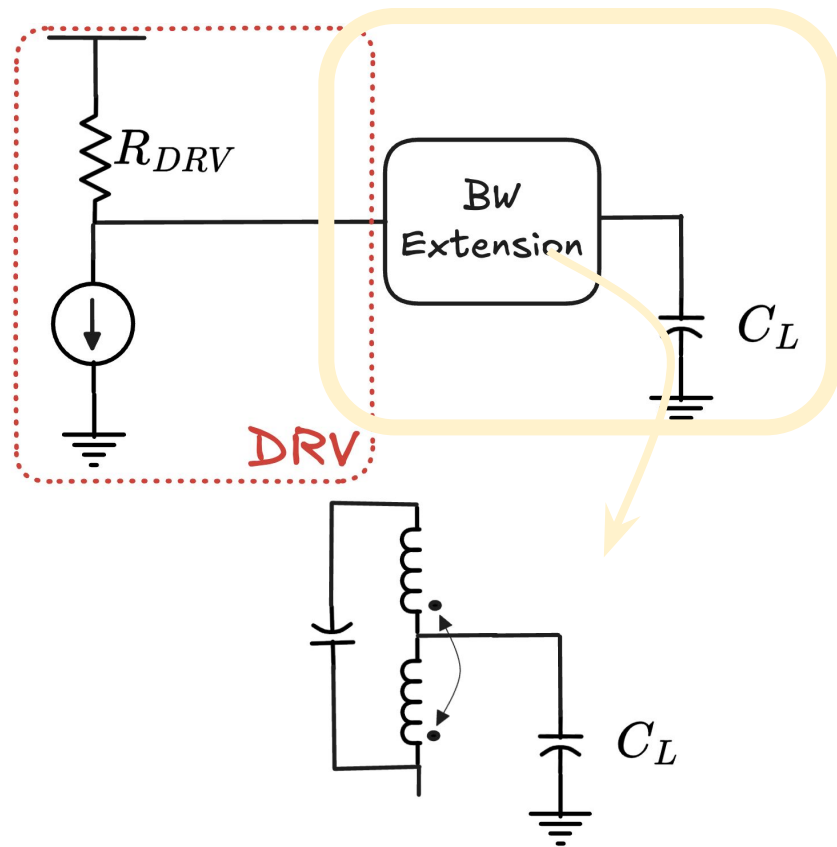
$$f_{3dB} = \text{BW Extension Ratio} \times \frac{1}{2\pi R_{DRV} C_L}$$

Targeting 3dB BW of ~70GHz.

A T-coil can be used to extend the BW.

Assuming a 2.8X BW extension for T-coil, the termination resistance can be calculated to be:

$$R_{DRV} \leq 42\Omega$$



Fundamental Limitations: Current Mode pJ/bit

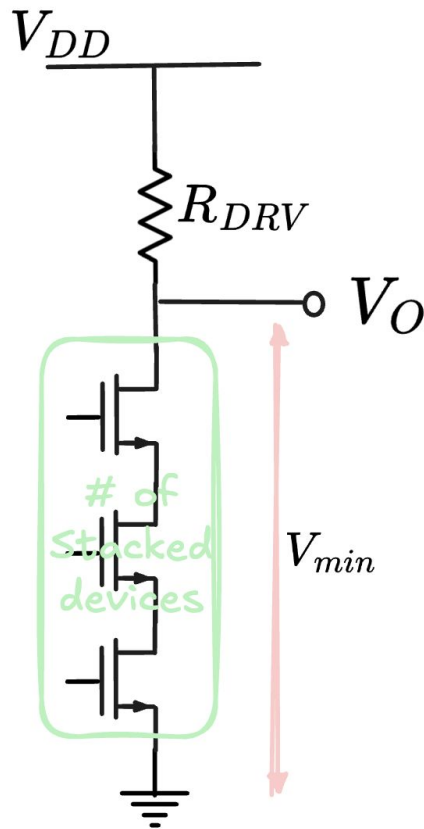
$$V_{DD} \geq V_{min} + V_{swing,pp} \quad I_{avg} = \frac{V_{DD} - V_{O,avg}}{R_{DRV}}$$

$$V_{O,avg} = V_{min} + \frac{V_{swing,pp}}{2} \quad P_{avg} = V_{DD} I_{avg}$$

Swing (pp,diff)	No stacks	Vmin	Vdd_min	Vo_DC	Iavg (mA)	Pavg	pJ/bit (diff)
2.00	2.00	0.20	1.20	0.70	11.11	13.33	0.27
3.00	2.00	0.20	1.70	0.95	16.67	28.33	0.57

Note: This energy calculation is ONLY for the output stage and does not include pre-driver stages, etc.

Thanks to Danyang for the discussions on the Current Mode driver



Fundamental Limitations: Voltage Mode

$$f_{3dB} = \text{BW Extension Ratio} \times \frac{1}{2\pi R_{DRV} C_L}$$

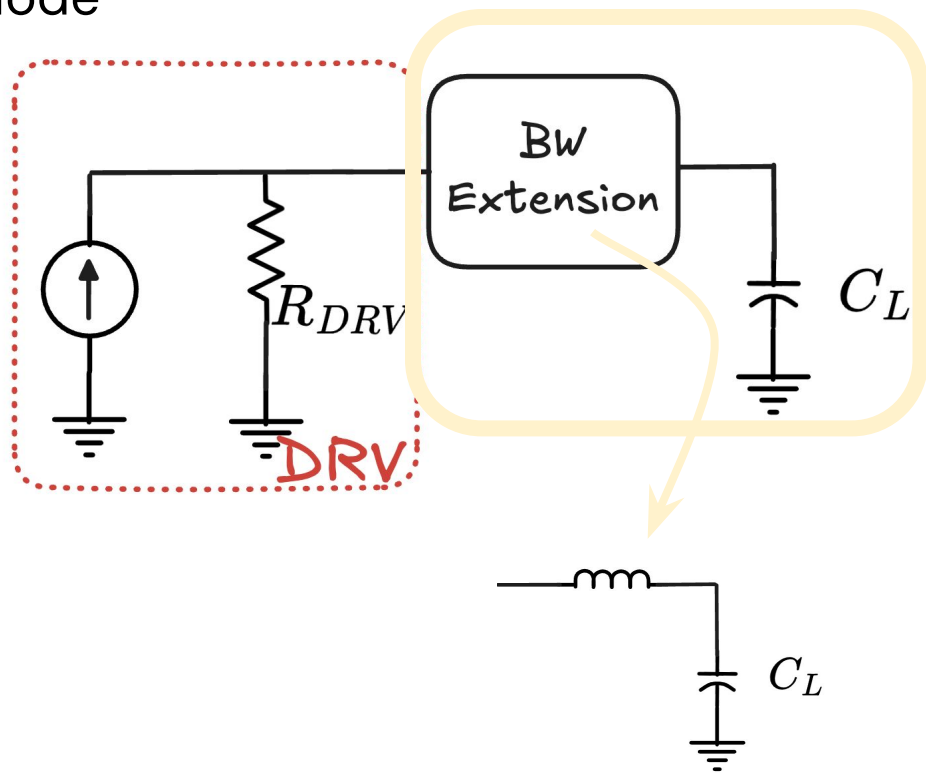
Targeting 3dB BW of ~70GHz.

Due to non-matched nature of this output stage, T-coil BW extension is not used..

Chose series inductive peaking.

The BW extension ratio is ~1.4X, which dictates the driver output impedance:

$$R_{DRV} \leq 20\Omega$$

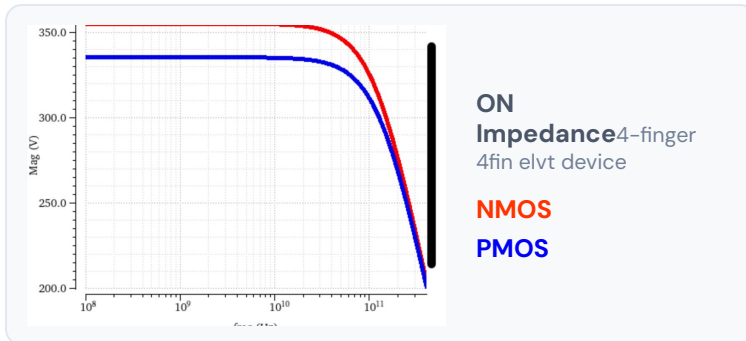


Fundamental Limitations: Voltage Mode pJ/bit

Target: 20 Ω ON Resistance

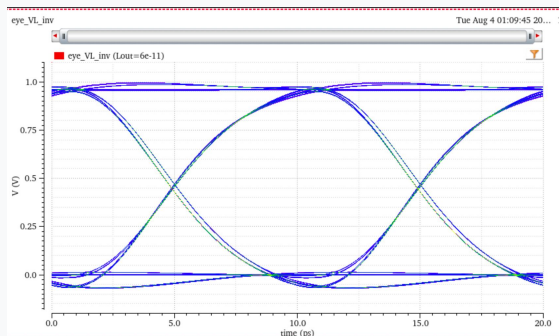
To achieve this target, the driver finger counts must scale with the voltage swing. Below are the required configurations and simulated eye diagrams.

Voltage mode driver obtains better energy efficiency compared to current mode driver.



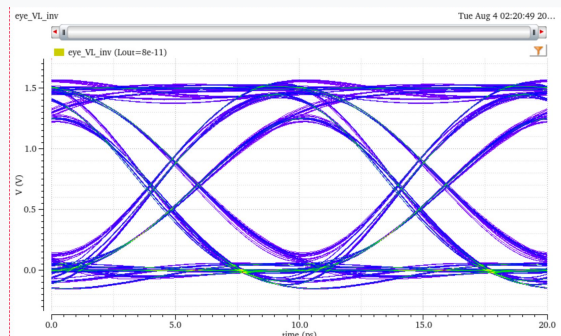
2V Swing, 0.1pJ/bit (diff)

Requires: 72 fingers PMOS/NMOS



3V Swing, 0.26pJ/bit (diff)

Requires: 2 X 144 fingers stacked PMOS/NMOS



A Note about pre-drivers

Current Mode

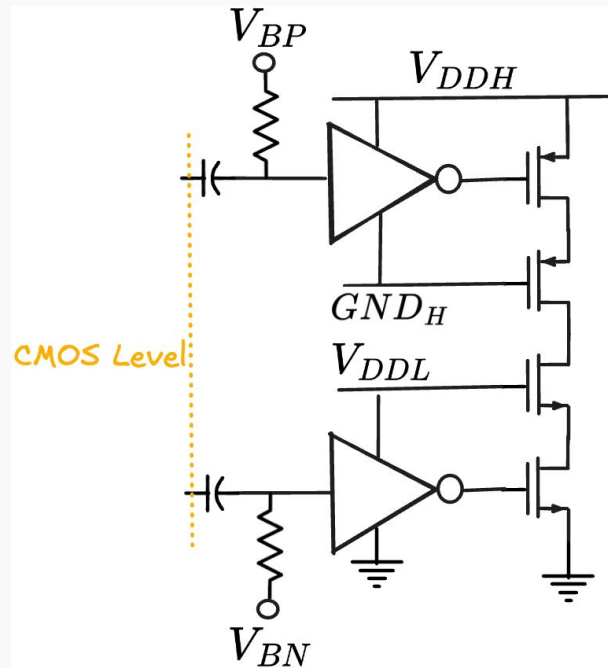
- Does not need an explicit driver stage.
- Can be driven directly using CMOS logic levels.

Voltage Mode

- **2V Swing:** Does not need an explicit driver.
- **3V Swing:** Requires separate pre-drivers for PMOS and NMOS.
- Pre-driver power consumption is **~2X** for high swing voltage mode driver compared to CML or low swing Voltage Mode.

* Note: The power consumption of the pre-driver is not included in the previous analysis.

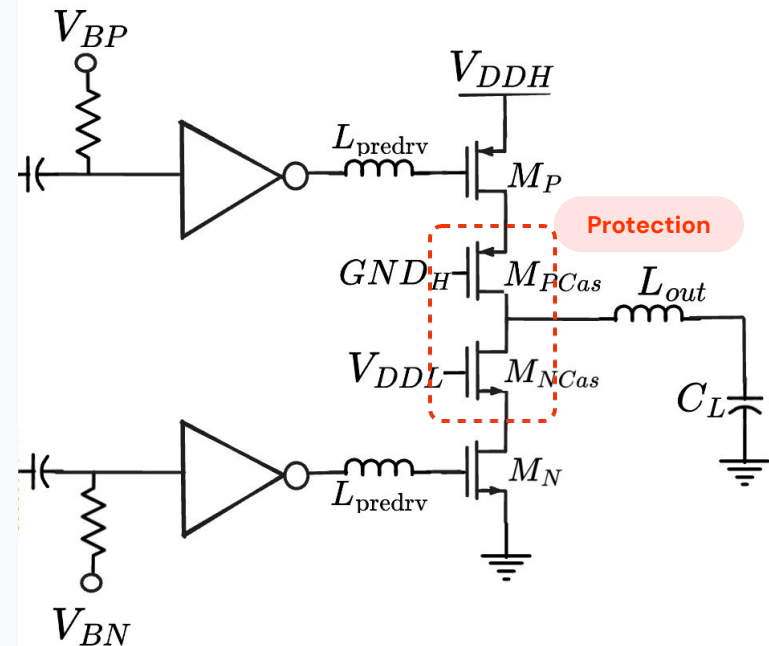
Conclusion: Since the high voltage mode driver output stage is **~2X more efficient** than the Current Mode driver, **voltage mode architecture is chosen** since the extra pre-driver power penalty would still be lower.



3V pp diff Voltage Mode Driver

Full Driver Topology: 3V Voltage Mode

Parameter	Value
VDDH	1.5V
VDDL	0.96V
GNDH	0.54V
Lout	80pH
MN, Mncas / MP, MPCas	120 finger/4fins each
VBP, VBN	<i>Tipping point of the inverters</i>
Lpredrv	<i>optimized for max eye opening based on predrv sizing</i>

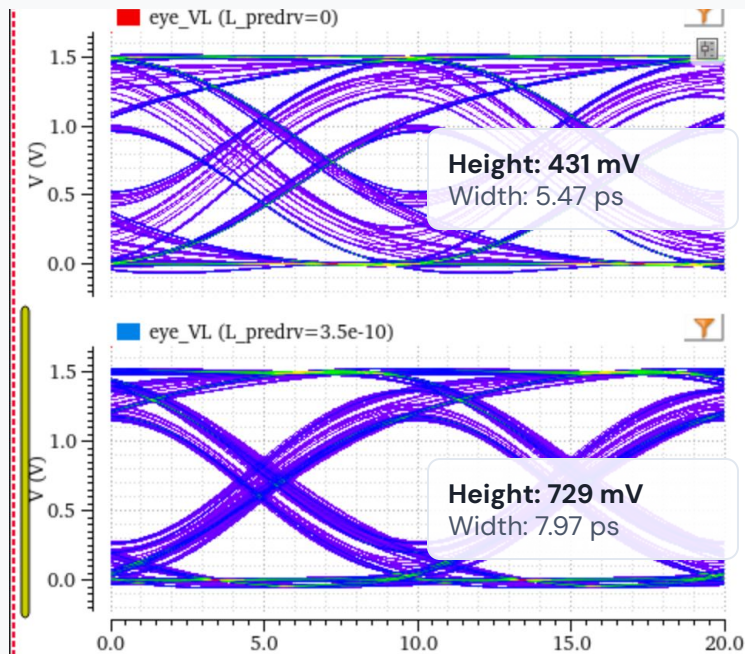


PreDriver design choices

** Note: Pre-driver peaking is adjusted to ensure maximum eye opening for each pre-driver sizing.*

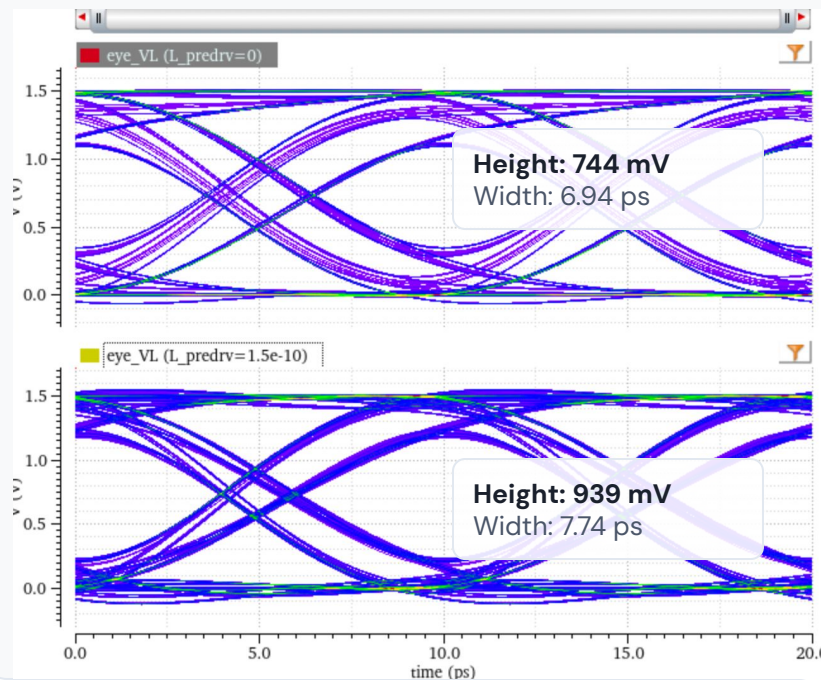
FO 4 Pre-Driver 330 fJ/bit (diff)

33% of Power in Predriver



FO 2 Pre-Driver 400 fJ/bit (diff)

40% of Power in Predriver



Output stage enhancement

AT OUTPUT MID-LEVELS

- Output devices operate in **Saturation Region**
- The circuit exhibits a **high gain**
- This high gain comes at a **bandwidth (BW) penalty**

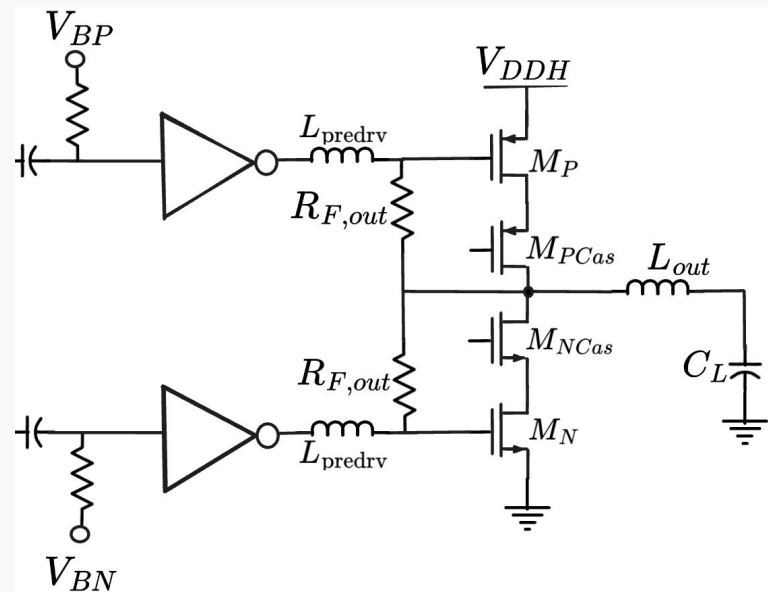
CLOSE TO OUTPUT SATURATION LEVELS

- Devices enter **Triode Region**
- Gain is **low**
- Bandwidth (BW) is **high**

DESIGN SOLUTION

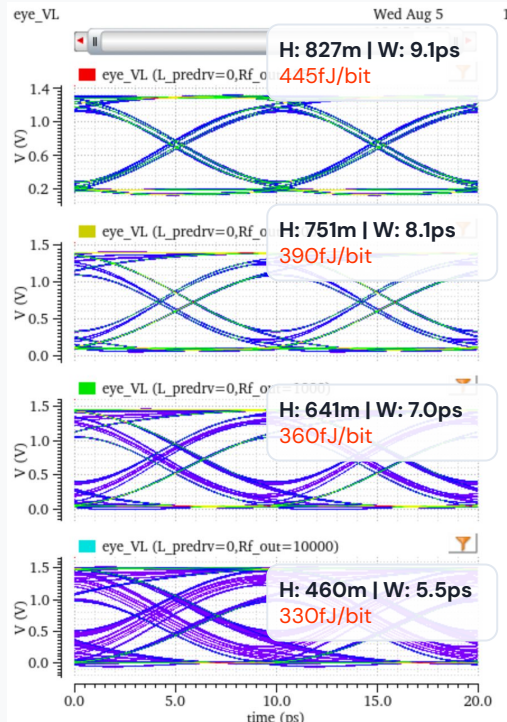
Avoid High Gain at output mid-levels.

Employing **resistive feedback** at the output stage can effectively enhance the BW.



Output stage with Resistive FeedBack

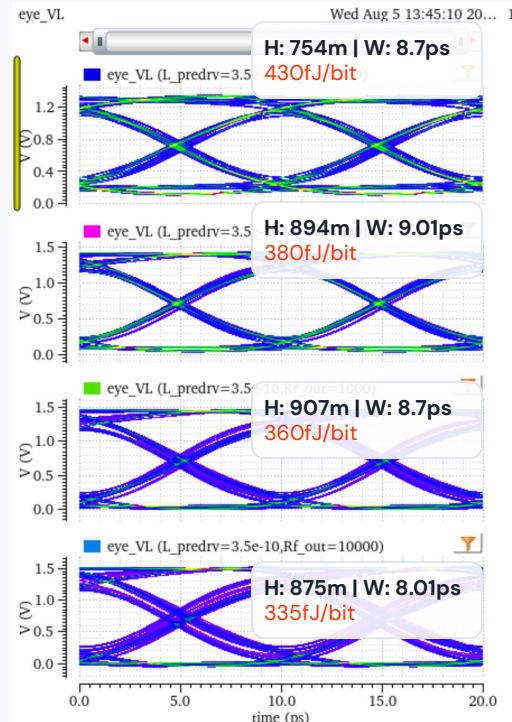
Lpredrv = 0



Lpredrv = 100pH



Lpredrv = 350pH



* Note: Obtained using a FO 4 pre-driver.

Output stage with Resistive FeedBack: Summary

PERFORMANCE & POWER PENALTY

Resistive feedback improves the BW, but adds a power penalty of **~100fJ/bit**.

EVALUATION VS. EQUAL-POWER APPROACHES

- Increase the sizing of the output stage
- Increase the sizing of the pre-driver

Note: The feedback resistor approach adds headroom cost since the output can't swing fully to VDD/GND.

KEY FINDING

The eye is more well-behaved for resistive feedback but suffers from **reduced swing at the output (25-15% smaller depending on Rf)**.

This is the root cause for lower eye-height in this method compared to the other two.

Pre-driv FO 1.5

H: 1025mV
W: 7.2ps

Pre-driv FO 2

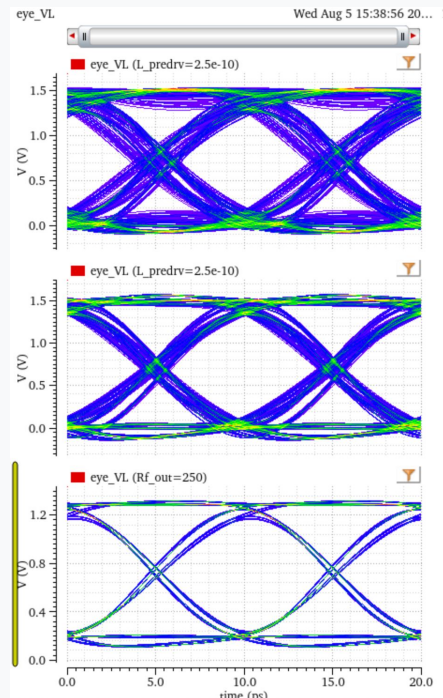
Output 13% bigger

H: 1180mV
W: 8.2ps

Pre-driv FO 4

250Ω resistive FB

H: 924mV
W: 9.2ps

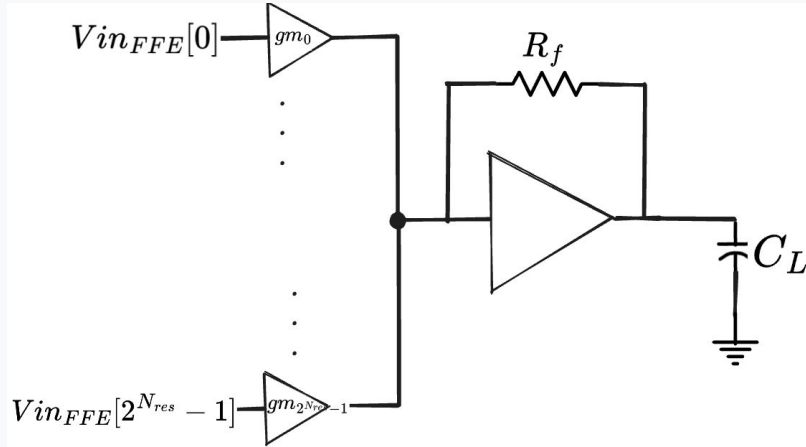


Power Consumption: All ~440fJ/bit (diff)

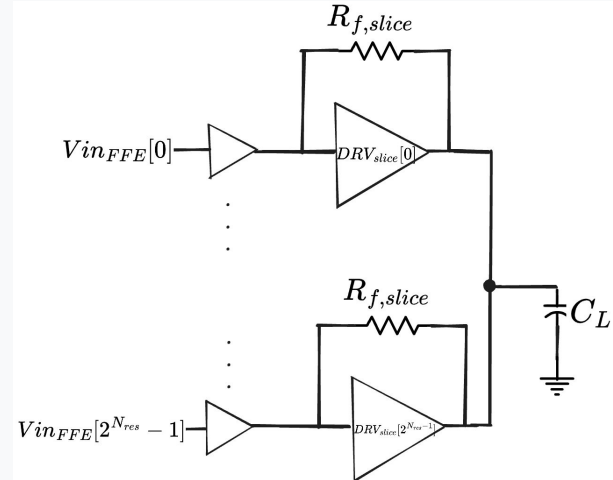
Implementation of Analog FIR: Two approaches

Performing the summation in current mode OR voltage mode.

Current mode FIR



Voltage mode FIR



Implementation of Analog FIR: Two approaches

Both of the implementations are evaluated at a high level.

Current Mode Implementation

- Adds power overhead due to voltage contention at the output
- Very tight balance between pre-driver speed and DAC sensitivity
 - Eye closure penalty is severe

Voltage Mode Implementation

- Non-linear behavior ($INL > 1LSB$)
- Linearization requires additional added resistance and increases power consumption
- Also adds power overhead, but less than the Current mode implementation.
- Requires very large resistors

Common Limitation

Both of these methods interfere with the main signal path and add additional loading for muxing/input selection between main/pre/post taps.

True Analog FIR implementation method

Implementation Method

Given the small coefficients currently used, a full DAC implementation is not required.

- Adds a pre/post dependent fraction of main tap current to the output node.
- Thick oxide devices in Cascode safely turn OFF unused taps.
- No major effects on the main path (only small parasitic routing added).

Risk & Mitigation

Asymmetry Risk

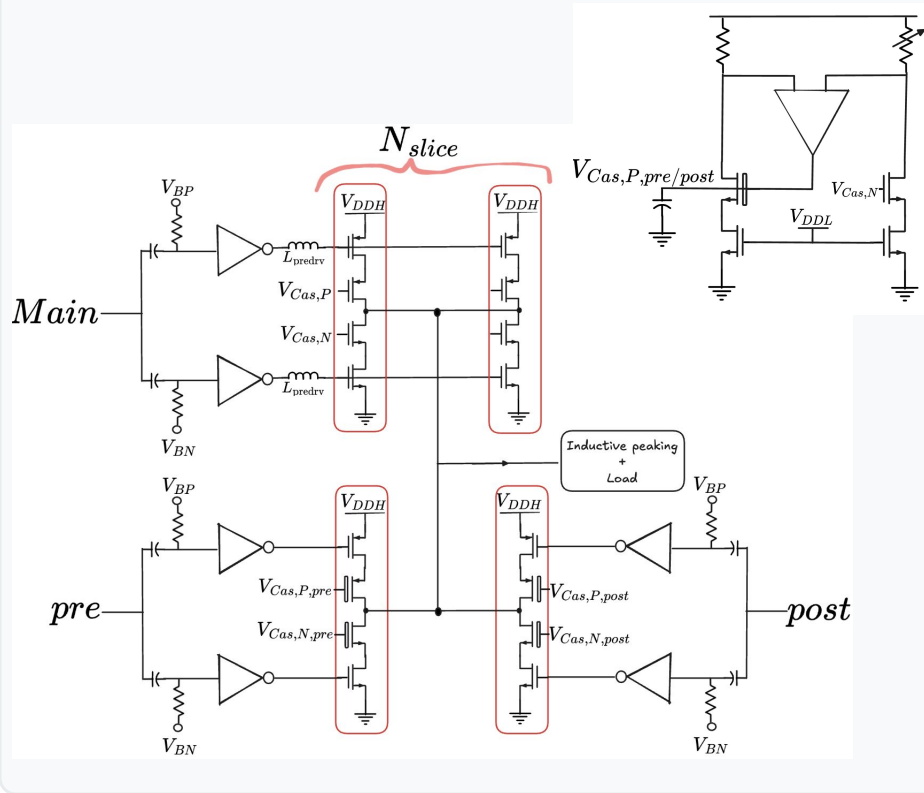
Inherent asymmetry between main path (lacks inductors) and pre/post paths.

Proposed Mitigation

Mitigated by introducing an intentional delay between main and pre/post path.

Question: Would it be possible to get pre/post data slightly (~1 gate delay) earlier than the main?

Is there a possibility to tune/select this delay



Input Loading of the Driver

Configuration	Fan-Out	Devices	Parasitic Routing	Total Capacitance
3V Swing	FO 2	86.4 fF	43.4 fF	129.8 fF
3V Swing	FO 4	43.2 fF	36.2 fF	79.4 fF
2V Swing	FO 2	21.6 fF	27.6 fF	49.2 fF
2V Swing	FO 4	10.8 fF	25.8 fF	36.6 fF

Loading Reduction Possibility: Device related portion of the parasitic routing of the 3V driver can be reduced by employing additional buffering stages, but at a bandwidth and power penalty.

Supply Requirements

3V Swing Requirements

Main Supply (Signal Path)

> 1.5 V

MRM Bias

> 1.7 V

Output ESD Protection

> MRM Bias + 0.75 V

2V Swing Requirements

Main Supply (Signal Path)

Between 0.9V and 0.96V

MRM Bias

> 1.2 V

Output ESD Protection

> MRM Bias + 0.5 V

* **Lower Swing Benefit:** Stems from avoiding device stacking. A single device can handle up to 0.96V and therefore the supply should be close to this level but not exceeding it.

** **MRM Bias Level:** Assumes a 0.2V drop on the resistor due to photodiode current to avoid forward biasing the MRM. The actual bias voltage needed will be determined later.

- *Different biasing technique may lower the needed supply for the MRM bias circuitry and enable sharing the supply between MRM bias and the main supply.*

Questions for SerDes Team



01 / ARCHITECTURE

SerDes Output

Architecture: Muxing scheme (2:1/4:1/8:1), differential vs. single ended output, drive impedance

Characteristics: Common Mode, Amplitude, jitter., possibility of getting a SerDes output waveform, etc.



02 / CHARACTERISTICS

Available Resources

Bandgap, reference currents, voltages



03 / CAPABILITY

SerDes Output Driving Capability

Analysis of serdes driving capabilities and scaling opportunities.



04 / TAPS & TIMING

Availability of Pre/Post Taps

Analysis of tap options availability and their respective timing